

## Revision · Probability

Cambridge insert: putting a number on “how likely”, and combining two events.

LESSON 99–100

2 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 12.1–12.4

### LESSON CLOCK



Order five events by likelihood 6 min

The basic formula 10 min

Complement and mutually exclusive events 10 min

Two events 10 min

Homework 4 min

### LEARNING OBJECTIVES

- Calculate the probability of an event from equally likely outcomes.
- Use the fact that probabilities of an event and its complement add to 1.
- Use a table or tree diagram for two events.
- Compare experimental with theoretical probability.

### TEXTBOOK

|           |                               |
|-----------|-------------------------------|
| National  | Annual revision               |
| Cambridge | Learner's Book<br>pp. 255–278 |
| Workbook  | Workbook 12.1–<br>12.4        |

### 01 Explanation

#### Putting a number on chance

##### FOR EQUALLY LIKELY OUTCOMES

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Every probability lies between 0 (impossible) and 1 (certain).

One die:  $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$ ,  $P(7) = 0$ ,  $P(\text{less than } 7) = 1$ . Give the answer as a fraction in its simplest form unless a decimal or percentage is asked for.

The word “equally likely” is doing real work. A drawing pin can land point-up or point-down, but not with probability  $\frac{1}{2}$  each — only an experiment can tell you.

#### Complement, and events that exclude each other

$$P(\text{not } A) = 1 - P(A)$$

If the probability of rain is  $0.3$ , the probability of no rain is  $0.7$ . This one line turns many “at least one” questions into a single subtraction — exactly as in the counting lesson.

#### MUTUALLY EXCLUSIVE EVENTS

If  $A$  and  $B$  cannot happen together,  $P(A \text{ or } B) = P(A) + P(B)$ .

A die shows a 2 or a 5:  $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$ .

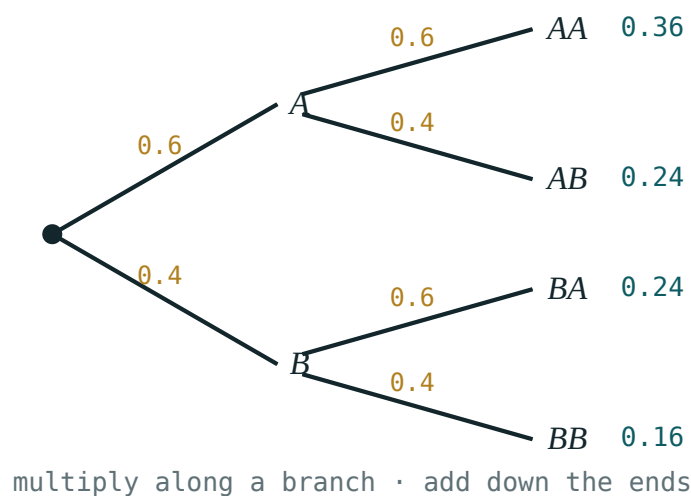
#### ONLY WHEN THEY EXCLUDE EACH OTHER

“Even” and “greater than 3” are not mutually exclusive — 4 and 6 belong to both — so adding the probabilities double-counts them.

## Two events, and what experiments show

For two **independent** events, multiply along the branches of a tree:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$



*Multiply along a branch to get the probability of that path; add the paths that satisfy the event.*

Two coins:  $P(\text{two heads}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ . For “exactly one head” there are two successful paths, HT and TH, so add:  $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ .

|            | THEORETICAL                              | EXPERIMENTAL                                             |
|------------|------------------------------------------|----------------------------------------------------------|
| Comes from | counting outcomes                        | actually doing the trials                                |
| Formula    | $\frac{\text{favourable}}{\text{total}}$ | $\frac{\text{times it happened}}{\text{times tried}}$    |
| Behaviour  | fixed                                    | settles towards the theoretical value as trials increase |

**Expected frequency** = probability × number of trials. In 300 throws of a die you expect about  $300 \cdot \frac{1}{6} = 50$  sixes — “about”, never exactly.

02

## Worked examples

EXAMPLE  
1

A bag holds 5 red, 3 blue and 4 green counters. One is taken at random. Find  $P(\text{red})$  and  $P(\text{not green})$ .

1

$5 + 3 + 4 = 12$   
Total outcomes.

2

$P(\text{red}) = \frac{5}{12}$

3

$P(\text{green}) = \frac{4}{12} = \frac{1}{3}$

4

$P(\text{not green}) = 1 - \frac{1}{3} = \frac{2}{3}$   
Use the complement.

ANSWER

$P(\text{red}) = \frac{5}{12}, P(\text{not green}) = \frac{2}{3}$

**EXAMPLE 2** Two dice are thrown. Find the probability that the total is 9.

- ① 36 equally likely outcomes  
A 6 by 6 table.

- ② (3,6), (4,5), (5,4), (6,3)  
Four ways to make 9.

- ③  $P = \frac{4}{36} = \frac{1}{9}$

ANSWER

$$\frac{1}{9}$$

**EXAMPLE 3** The probability that a bus is late is 0.2. Find the probability that it is late on both of two days, and that it is late on exactly one.

- ①  $P(\text{late, late}) = 0.2 \cdot 0.2 = 0.04$   
Independent days — multiply.

- ②  $P(\text{late, on time}) = 0.2 \cdot 0.8 = 0.16$   
First branch.

- ③  $P(\text{on time, late}) = 0.8 \cdot 0.2 = 0.16$   
Second branch.

- ④  $0.16 + 0.16 = 0.32$   
Add the two successful paths.

ANSWER

0.04 and 0.32

### 03 Interactive model

Work through one probability question at a time; ask for the answer before revealing each step.

#### Probability, step by step

A card numbered 1 to 20 is drawn. Find  $P(\text{multiple of } 3)$ .

① 3, 6, 9, 12, 15, 18

List  
the  
favourable  
outcomes.

② 6

favourable  
out  
of

③  $\frac{6}{20} = \frac{3}{10}$

Simplify.

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                  | O'ZBEKCHA                   | РУССКИЙ                         |
|--------------------------|-----------------------------|---------------------------------|
| Probability              | Ehtimollik                  | Вероятность                     |
| Event                    | Hodisa                      | Событие                         |
| Equally likely           | Teng imkoniyatli            | Равновозможные                  |
| Favourable outcome       | Qulay natija                | Благоприятный исход             |
| Complement               | Qarama-qarshi hodisa        | Противоположное событие         |
| Mutually exclusive       | Birgalikda bo'lmaydigan     | Несовместные                    |
| Independent events       | Bog'liq bo'lmagan hodisalar | Независимые события             |
| Tree diagram             | Daraxt diagrammasi          | Дерево вероятностей             |
| Experimental probability | Tajribaviy ehtimollik       | Опытная (частотная) вероятность |
| Expected frequency       | Kutilayotgan chastota       | Ожидаемая частота               |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A die is thrown.  $P(\text{more than } 4)$  is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

If  $P(A) = 0.42$  then  $P(\text{not } A)$  is:

$$0.42$$

$$0.58$$

$$0.68$$

$$1.42$$

Two coins are tossed.  $P(\text{exactly one head})$  is:

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{3}{4}$$

A spinner has  $P(\text{blue}) = 0.25$ . In 80 spins expect about:

16

20

25

40

Which pair is NOT mutually exclusive for one die?

“2” and “5”

“even” and “odd”

“even” and “more than 3”

“1” and “6”

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.



## Easy · warm the idea up

7 PROBLEMS

1. A die is thrown. Find  $P(3)$ .

ANSWER  $\frac{1}{6}$

2. A die is thrown. Find  $P(\text{odd})$ .

ANSWER  $\frac{1}{2}$

3. A bag has 3 red and 7 blue. Find  $P(\text{red})$ .

ANSWER  $\frac{3}{10}$

4. If  $P(\text{win}) = 0.4$ , find  $P(\text{not win})$ .

ANSWER 0.6

5. A coin is tossed twice. Find  $P(\text{two tails})$ .

ANSWER  $\frac{1}{4}$

6. A card is drawn from 1–10. Find  $P(\text{more than 7})$ .

ANSWER  $\frac{3}{10}$

7. A spinner has  $P(\text{red}) = 0.5$ . In 60 spins, expect how many reds?

ANSWER 30

## Medium · the standard question

7 PROBLEMS

1. *Two dice are thrown. Find  $P(\text{total } 7)$ .*

ANSWER  $\frac{1}{6}$

2. *A bag has 4 red, 5 blue, 6 green. Find  $P(\text{not blue})$ .*

ANSWER  $\frac{2}{3}$

3.  *$P(\text{late}) = 0.15$ . Find  $P(\text{late on both of two days})$ .*

ANSWER 0.0225

4. *A coin is tossed three times. Find  $P(\text{exactly two heads})$ .*

ANSWER  $\frac{3}{8}$

5. *A card is drawn from 1–20. Find  $P(\text{prime})$ .*

ANSWER  $\frac{8}{20} = \frac{2}{5}$

6.  *$P(A) = 0.3$ ,  $P(B) = 0.5$  and  $A, B$  are mutually exclusive. Find  $P(A \text{ or } B)$ .*

ANSWER 0.8

7. *A spinner has  $P(\text{green}) = 0.35$ . In 400 spins, expect how many greens?*

ANSWER 140

**Hard · stretch and reason**

7 PROBLEMS

1. A bag has 5 red and 4 blue. Two are taken without replacement. Find  $P(\text{both red})$ .

ANSWER  $\frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$

2. Two dice are thrown. Find  $P(\text{at least one 6})$ .

ANSWER  $\frac{11}{36}$

3.  $P(\text{rain}) = 0.4$  on each of three days. Find  $P(\text{no rain at all})$ .

ANSWER  $0.6^3 = 0.216$

4.  $P(\text{rain}) = 0.4$  on each of three days. Find  $P(\text{at least one wet day})$ .

ANSWER  $1 - 0.216 = 0.784$

5. A bag has 6 red and 4 blue. One is taken and replaced, then another. Find  $P(\text{different colours})$ .

ANSWER  $2 \cdot 0.6 \cdot 0.4 = 0.48$

6. A die is thrown 120 times and gives 32 sixes. Compare experimental and theoretical probability.

ANSWER  $\frac{32}{120} \approx 0.27$  against  $\frac{1}{6} \approx 0.17$  — noticeably high; either chance, or the die is biased

7. Explain why  $P(A \text{ or } B) = P(A) + P(B)$  can fail.

ANSWER If A and B can happen together, the outcomes in both are counted twice; the correct rule subtracts  $P(A \text{ and } B)$ .

## 07 Homework

### Homework — 6 problems

Cambridge Stage 9, Unit 12. Simplify every fraction.

- A bag has 7 red and 5 blue. Find  $P(\text{blue})$  and  $P(\text{not blue})$ .
- Two dice are thrown. Find  $P(\text{total 5})$ .

3. A coin is tossed three times. Find  $P(\text{at least one tail})$ .
4.  $P(\text{faulty}) = 0.05$ . In 600 items, how many faulty ones are expected?
5. A card is drawn from 1–15. Find  $P(\text{multiple of 4})$ .
6. Explain the difference between experimental and theoretical probability in two sentences.